

An infrared triangle for quantum spin chains

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The infrared triangle—soft theorem, asymptotic symmetry, memory effect—comes to quantum spin chains. Proved: a truncated symmetry acts only at its endpoints, its charge algebra centrally extended by the topological index; in the bilinear isotropic ferromagnet—regular two-magnon domain, fixed hard momentum inside the band, unit charge—the soft phase slope is universal; granted a scattering hypothesis, a quantized wall step per transmitted magnon. Its edges stay conjectures. The continuum rule, memory as zero-frequency soft factor, fails: the soft factor is a phase, the memory a charge.

A ferromagnetic spin chain contains two objects that can be pictured without formalism: the *magnon*, a single overturned spin spreading as a wave, and the *domain wall*, the boundary between a region of up spins and a region of down spins. This Letter concerns what each does as the magnon’s momentum is taken small, and the bookkeeping that controls both. The whole result in one sentence: in an exactly solved two-magnon problem a soft magnon exits with a universal phase slope (Theorem 1); the symmetry behind this acts at the ends of the chain as a concrete algebra of operators on matrix-product bond space; granted the scattering hypothesis H-AD-G stated below, a magnon that crosses a domain wall displaces it by a quantized charge step—and the continuum rule that identifies the memory with the zero-frequency limit of the soft factor is *false* here: what survives, and is proved, is a current-flux identity and charge bookkeeping—charge transport, not phase accumulation.

In continuum field theory, soft theorems, asymptotic symmetries, and memory effects form the *infrared triangle*, three faces of one Ward identity, established for photons and gravitons at null infinity [1, 2]. Hamada and Sugishita extended the triangle to Goldstone bosons of a broken global symmetry [3]; scalar and fracton versions followed [4–6], with local derivations avoiding null infinity [7] and infrared finiteness [8]. Each corner has a condensed-matter ancestor: soft-magnon decoupling and Adler zeros of nonrelativistic Goldstone theories are by now systematic [9–12]; Lan and Xiao described magnon-driven domain-wall displacement as a two-body collision [13, 14]. Missing is a lattice statement at finite spacing, no continuum limit; we supply one, prove what we can, label the rest conjecture, and report one refutation as a central result.

Setting.—An infinite chain of finite-dimensional sites; a compact symmetry group G acting on-site by $u(g)$; a finite-range G -invariant Hamiltonian H ; and a family of translation-invariant injective matrix-product ground states with tensors A_α and bond dimension χ , one for each vacuum α , permuted by the symmetry as $\alpha \mapsto g \cdot \alpha$. One structural fact is used throughout, the fundamental theorem of MPS symmetries [15, 16]: acting with $u(g)$ on the physical index of an invariant A_α equals a phase

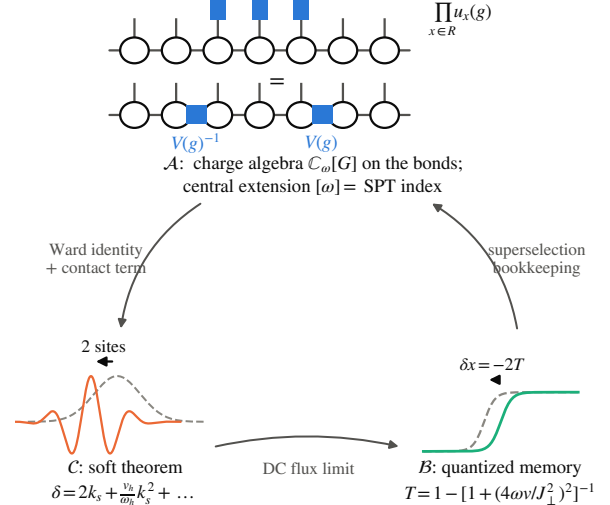


FIG. 1. The infrared triangle at finite lattice spacing. Corner $\mathcal{A}$ (asymptotic symmetry, proved): a symmetry applied to a region of an MPS collapses to two bond operators $V(g)^{-1}, V(g)$ at the region’s ends; on windows padded about a bond these generate the charge algebra $\mathbb{C}_\omega[G]$, central extension $[\omega]$ the SPT index. Corner $\mathcal{C}$ (soft theorem, proved for two-body scattering in the bilinear isotropic ferromagnet on its regular two-magnon domain—fixed hard momentum inside $(0, \pi)$, fixed channel, unit charge; band edges and equal velocities excluded): a soft magnon exits with phase slope 2 at spin $\frac{1}{2}$, i.e. displaced by two sites ($1/s$ at site spin s). Corner $\mathcal{B}$ (memory, proved conditional on the scattering hypothesis H-AD-G: complete kink–magnon wave operators for a fixed packet with no bound component, one reflected and one transmitted channel of definite charges ∓ 1 , local decay, limits taken in the order volume, time, window): a transmitted magnon displaces the wall by exactly $-1/s$ sites, a reflected one by zero. No edge is a theorem: the $\mathcal{A} \Rightarrow \mathcal{C}$ edge remains Conjecture S at every order—Corner $\mathcal{C}$ itself is proved two-body; $\mathcal{C} \Rightarrow \mathcal{B}$ passes soft data to memory only through the transmission probability; $\mathcal{B} \Rightarrow \mathcal{A}$ is charge bookkeeping within a fixed kink sector.

times $V(g)^{-1} A_\alpha V(g)$, with V a projective bond representation whose class $[\omega] \in H^2(G, U(1))$ is the symmetry-protected topological (SPT) index [15, 17]. All theorems below are stated in this $(G, \text{injective MPS, finite range})$

register; Appendixes A–D linearize the proofs, with the repository’s shards and machine-checked certificates as ground truth (Appendix E).

Asymptotic symmetry.—Apply the symmetry to a finite region R only: $U_R(g) = \prod_{x \in R} u_x(g)$. Used once per site, the intertwining relation cancels the V ’s pairwise through the interior: the operation collapses to two bond insertions, $V(g)^{-1}$ entering R and $V(g)$ leaving it (Fig. 1). Everything a truncated symmetry does, it does at its two ends. Sending an end to infinity makes this an asymptotic-symmetry statement; three facts hold (Appendix A). (i) On states, a half-infinite symmetry string acts *exactly* as a single bond insertion; as an operator sequence it converges strongly if and only if $V(g)$ is a phase—otherwise there is no strongly convergent implementer at all, the lattice counterpart of a large gauge transformation. (ii) On windows padded around the distinguished bond the insertions compose as the twisted group algebra $\mathbb{C}_\omega[G]$: the SPT class is precisely the central extension of the asymptotic charge algebra—a group-cohomological multiplier, not a Lie-algebra central charge (Appendix D). Whether these window states extend to normal states of the physical half-infinite algebra is open—hypothesis H-split; no statement below uses it silently. (iii) If the symmetry is broken, the half-infinite string leaves the vacuum sector: every finite truncation remains in the vacuum folium (it carries a kink–antikink pair at its two ends), and only the limit—in expectation values, never in norm—lands in a *domain-wall* (kink) sector. Kinks are what broken asymptotic symmetries create. When G acts transitively on the vacua, kink sectors are labelled by ordered vacuum pairs, with double-coset invariant in $H_\alpha \backslash G / H_\alpha$ (H_α the unbroken subgroup).

The same telescoping yields a lattice Noether identity: on the vacuum, for a normal-ordered *unbroken* generator, the physical charge density equals the divergence of a purely virtual bond quantity; separately, for *any* generator and any finite-range invariant H the deformed charge Q_k obeys the exact continuity equation $[H, Q_k] = (e^{ik} - 1)J_k$. The kinematic factor is a property of the profile, not a soft factor; the soft theorem must be earned from the current.

The soft theorem.—Fix the concrete model: the spin- $\frac{1}{2}$ Heisenberg ferromagnet $H = -J \sum_x (\mathbf{S}_x \cdot \mathbf{S}_{x+1} - \frac{1}{4})$. Its polarized ground state is an exact $\chi = 1$ MPS, magnons are one-particle eigenstates with $\omega(k) = J(1 - \cos k)$ and velocity $v(k) = J \sin k$, and $SU(2)$ breaks to $U(1)$, making the magnon a type-B Goldstone mode [9]. Write $S_{12} = e^{i\delta}$ for the exact two-magnon amplitude, δ the continuous phase branch vanishing with the soft momentum k_s at fixed hard k_h .

Theorem 1 (soft magnon theorem, two-body). *For k_h in a compact subset of $(0, \pi)$ and signed $k_s \rightarrow 0$,*

$$\delta = 2 \operatorname{sgn}(v_h - v_s) k_s + \frac{|v_h|}{\omega_h} k_s^2 + O(k_s^3) \quad (1)$$

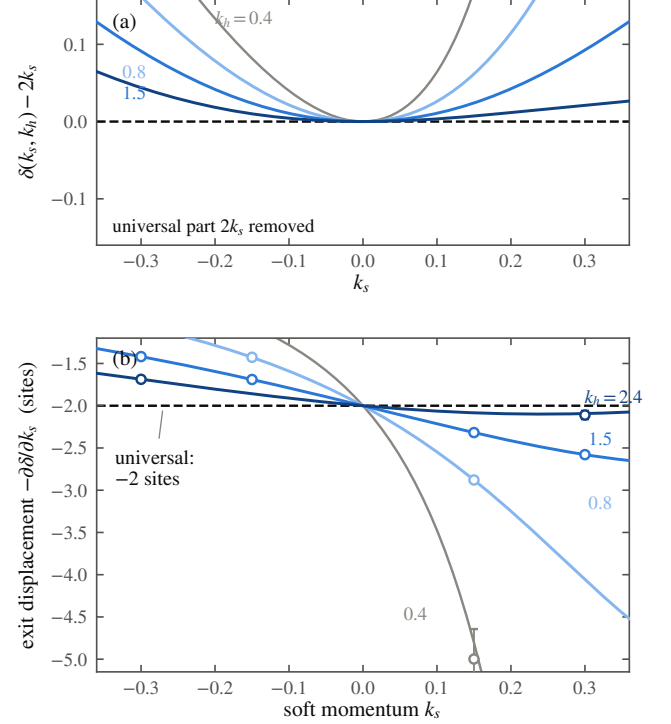


FIG. 2. The soft magnon theorem and its verification. (a) The exact two-magnon phase of the Heisenberg ferromagnet with its universal part removed: $\delta(k_s, k_h) - 2k_s$ vanishes quadratically for each hard momentum k_h in a compact subset of $(0, \pi)$ —the hard leg enters only at order k_s^2 , through the magnon frequency $\omega_h = J(1 - \cos k_h)$ and velocity $v_h = J \sin k_h$, with coefficient v_h/ω_h , Eq. (1). (b) The displacement reading: the soft packet’s exit displacement $-\partial\delta/\partial k_s$ for the same k_h range (curves, exact) passes through -2 sites at $k_s = 0$. Circles: real-time wavepacket measurements, which reproduce the universal coefficient 2 to 0.2%. The gray curve ($k_h = 0.4$) shows the non-uniformity of the expansion as $k_h \rightarrow 0$; the universal -2 survives it. Every soft magnon leaves a two-site footprint, whatever magnon it scatters off—within two-body scattering; beyond is Conjecture S, unproved.

uniformly, with $v_h = v(k_h)$, $\omega_h = \omega(k_h)$.

The zeroth order, $S_{12} \rightarrow 1$, is soft-magnon decoupling (a zero-momentum magnon is a symmetry rotation of the vacuum); the content is the next two orders. The linear coefficient is the pure number 2, up to the sign that records which particle is faster: independent of the hard momentum and of the coupling on the stated compact domain. Hard data enters first at second order, through the single even invariant $|v_h|/\omega_h$. And the 2 is not a scattering length—the ferromagnet’s scattering length vanishes [12]—it is a *displacement*: the soft packet exits the collision shifted by two lattice sites (Fig. 2). The derivation (Appendix B) does not solve the model: one contact relation plus the implicit function theorem fixes the linear

coefficient, all hard dependence cancelling. It has a Ward reading: the residue $\langle k_h | Q_0^\dagger J_0^- | k_h \rangle = 2i v_h$ is charge times velocity, mirroring the continuum soft-current analysis [3, 10]. Bethe integrability enters only as an oracle: the closed-form expansion reproduces Eq. (1) term by term, and a directly diagonalized two-magnon completeness theorem—not assumed—makes the expansion unconditional. Exact diagonalization confirms the form factors to 2×10^{-14} and both coefficients to 3×10^{-10} ; wavepacket collisions reproduce the displacement, the coefficient 2 to 0.2%.

How far beyond two bodies does this reach? The strongest objection first: universality *fails* for unrestricted local sources—an explicit four-site operator O_η creates the same one-magnon state yet shifts the linear soft coefficient by $2i\eta(1 - e^{-3ik_h})$ (Appendix B). We conjecture—Conjecture S—that for n -particle scattering on an injective MPS vacuum, with sources in the five-condition Ward-covariant no-contact class $\mathcal{S}_W$ of Appendix B, $M_{n+1} = S(k_s) M_n + o(k_s)$ in wavepacket norm, the multiplier S depending only on the legs’ asymptotic charges and velocities, linear at leading order. Theorem 1 is its proved two-body instance, $S = 2ik_s$. All else is open—wave operators, form-factor regularity, the soft-limit interchange, two or more hard legs, membership of any microscopic source in $\mathcal{S}_W$; Appendix B states the complete list.

Memory.—Memory needs a wall, so change the model minimally: the easy-axis XXZ ferromagnet, anisotropy $\Delta > 1$, with two product vacua and a domain wall between them; send in a magnon wavepacket and ask where the wall sits afterwards (Fig. 3). Read the wall position from a window W around it, $\mathfrak{X}_W = \text{const} + \frac{1}{2s} \sum_{x \in W} S_x^z$. Its change over the event is an exact operator identity, no hypotheses: the time-integrated $U(1)$ current through the two boundary bonds of W , i.e. the zero-frequency weight of the physical boundary current—as in the continuum, memory sits in the field at the boundary [2, 7]. Charge bookkeeping then gives the memory law, in the general register.

Theorem 2 (memory law). *Let G be compact, $\{A_\alpha\}$ a covariant family of injective MPS vacua of any bond dimension, H finite range and G -invariant; fix a kink sector between two vacua carrying charge density $\pm s$ along a common unbroken $U(1)$, and let the incoming state be one fixed kink-magnon wavepacket in that sector whose charge deviations from the two vacua are summable—the ℓ^1 kink class (no plane wave). Assume the scattering hypothesis H-AD-G (Appendix C) for this packet: wave operators exist and are complete on its sector; the outgoing space carries exactly one reflected and one transmitted channel, of charges $q_{\text{in}} = q_L = -1$ and $q_T = +1$ relative to their supporting vacua, and no further propagating channel; the packet has no bound-state component; local observables decay on the moving legs; and the limits are taken*

in the order infinite volume, then time, then observation window. Then the wall-displacement operator is

$$\Delta X = -\frac{1}{s} N_T, \quad \delta x = -\frac{1}{s} \langle N_T \rangle, \quad (2)$$

with N_T the transmitted-channel projection: each transmitted magnon displaces the wall by exactly $-1/s$ sites, each reflected one by zero, and $\text{spec}(\Delta X) \subseteq \{-1/s, 0\}$ with Bernoulli variance.

The displacement is quantized by charge conservation alone: momentum, anisotropy, packet shape, and scattering phase enter only through $\langle N_T \rangle$. The proof is charge bookkeeping (Appendix C). One hypothesis is structural: the two vacua must carry *opposite* charge density along one unbroken direction; the spin flip supplies this in XXZ, and a bare abelian symmetry cannot. H-AD-G itself we prove only for the dominant domain-wall sector of the XXZ chain, conditional on a stated sector-reduction hypothesis; for the full chain it remains an assumption (Appendix C). Sparse-sector numerics at $N = 160$ confirm $\delta x = -2 \langle N_T \rangle$ at $s = \frac{1}{2}$ to 0.064 sites across the scan (worst for the slowest packets, trapped weight $\approx 4 \times 10^{-3}$), and to 0.005 sites on the nine rows with trapped weight below 10^{-6} .

In the reduced sector the kink-magnon problem is a side-coupled (Fano) level, giving the closed form $t(k) = [1 + iJ^2/4\omega(k)v(k)]^{-1}$ with $\omega(k) = J(\Delta - \cos k)$; the measured reflection matches this over $\Delta \in [1.5, 12]$ to 0.9–5.8%, no fitted parameter (criterion 8%, fixed in advance). As $k \rightarrow 0$,

$$T(k) = 16(\Delta - 1)^2 k^2 + O(k^4), \quad (3)$$

so a soft magnon is totally reflected and leaves no memory: the memory effect has an Adler zero of its own. Whether the zero’s coefficient is universal we have not established; that remains a conjecture.

Now the refutation. The continuum dictionary—memory = DC limit of the soft factor, summed over the event—is *false* in the XXZ chain, twice over: the displacement per transmission is fixed by charge conservation whatever the scattering phase does, and a purely transmitting wall with identically zero phase shift still displaces by exactly $-1/s$. The soft factor is a phase; the memory quantum is a charge. What survives—and is proved—is the pair above: memory is the DC weight of the boundary current, and soft data reaches it only through the transmission probability: charge transport, not phase accumulation.

The remaining edge is bookkeeping, and only that: under finite-range dynamics the vacuum-pair label of the kink sector is rigid at finite times, and $2s\delta x + (q_{\text{out}} - q_{\text{in}}) = 0$ within that fixed sector. Whether the memory *reconstructs* the asymptotic transformation which acted remains open.

One invariant, read out twice.—Two 2’s appear above: the soft phase slope of Theorem 1 and the spin- $\frac{1}{2}$ memory quantum $1/s$ of Theorem 2. We conjecture—Conjecture B—that both equal $|q_{\text{hard}}|/s$, the hard particle’s $U(1)$ charge in units of the vacuum spin density. It was built to fail at $s \neq \frac{1}{2}$, where $1/s$ and the constant 2 separate; it survived (Fig. 3c, inset). Two independent implementations measured the phase slope = $1/s$ across $s \in \{\frac{1}{2}, 1, \frac{3}{2}, 2\}$ and the memory quantum = $-1/s$ across $s \in \{\frac{1}{2}, 1, \frac{3}{2}\}$, within pre-registered 8% bands (worst gated deviation 2.7%). An exact two-magnon computation now proves the slope half for the bilinear isotropic ferromagnet at every site spin: on the regular two-magnon domain, at each fixed hard momentum in $(0, \pi)$ and fixed channel, $\partial\delta/\partial k_s|_0 = \text{sgn}(v_h - v_s)/s$ in closed form, hard dependence cancelling (Appendix B)—the unit-charge two-body slope only, with band edges and equal velocities excluded. The memory half remains conditional on H-AD-G; the $|q_{\text{hard}}|$ factor is untested—every leg so far carries $|q| = 1$ —and a charge-2 bound-state projectile is the next falsifier.

Where the SPT index lives.—Corner $\mathcal{A}$ made $[\omega]$ the central extension of the soft charge algebra; does it show up in soft amplitudes? The answer, proved in Appendix D, is a rigidity dichotomy. In the *bulk*, a closed symmetry insertion has two endpoints whose projective multipliers cancel exactly, and every normalized bulk coefficient is continuous along common-gap symmetric deformation paths with continuously varying external data (a coefficient reached through a soft limit or p derivatives needs in addition C^p external data and the uniformity hypothesis H-soft- p , Appendix D): nothing bulk is a topological invariant without a separate local-constancy proof. Along a symmetric path through the AKLT phase a natural bulk soft-charge coefficient moves from 0.125 to 0.240, SPT class fixed. At an *unpaired endpoint* the situation inverts: the compensated soft residue is an operator on the χ -dimensional edge register whose centered charge has spectrum in the $[\omega]$ -shifted lattice $q_\omega^\circ + \mathbb{Z}$ and dimension at least the minimal ω -projective dimension. For the AKLT half-chain the residue is exactly $-\frac{1}{2}[1 - (2b^2 - 1)^L]Z \rightarrow -Z/2$ along the entire path: eigenvalues $\pm\frac{1}{2}$, rigid, while the bulk coefficient drifts. The topological data of soft physics concentrates at the boundary. Physical edge statements carry the hypothesis H-split above; granted in addition an edge-resolved scattering hypothesis H-AD-edge, charge conservation, and definite channel charges, charge bookkeeping quantizes edge-memory outcomes exactly as in Theorem 2: what is protected is the memory’s *capacity*—a projective multiplier no symmetric perturbation can split—not any particular amplitude. That an AKLT boundary magnon actually writes to it, we state as a conjecture with a definite Hamiltonian and a pre-registered test (Appendix D).

Outlook.—The chain’s topological data thus concentrates in a boundary soft algebra, $\mathbb{C}_\omega[G]$ at the ends, in

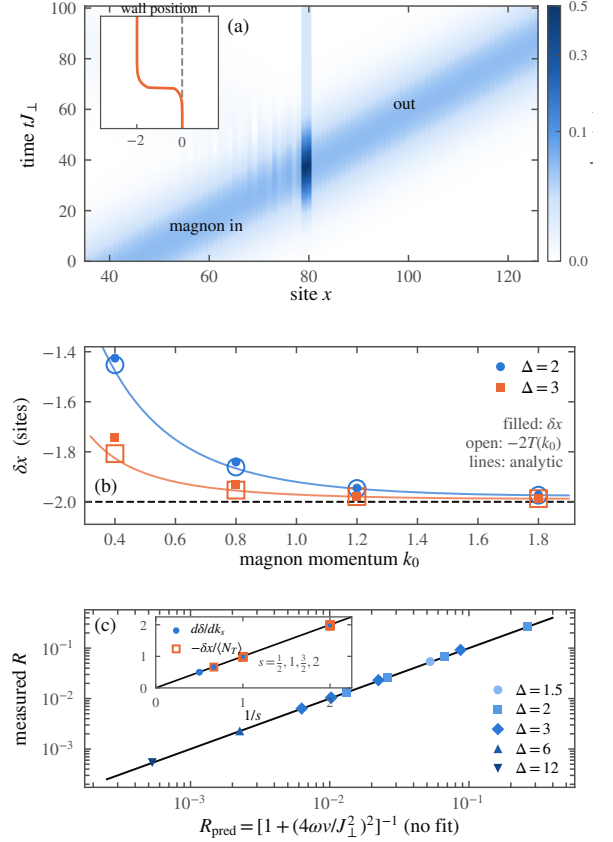


FIG. 3. The quantized memory effect in the XXZ chain ($\Delta > 1$). (a) Space-time record of one event ($\Delta = 3$, $k_0 = 1.2$, $N = 160$): flipped-spin density as a magnon wavepacket crosses the wall. Inset: the wall position steps back by two sites and stays—the permanent record. Across the plotted scan $\delta x = -2 \langle N_T \rangle$ ($\langle N_T \rangle$: transmitted weight) holds to 0.064 sites (worst: slowest packets, trapped weight $\approx 4 \times 10^{-3}$); on the nine rows with trapped weight $< 10^{-6}$, to 0.002 sites. (b) Displacement versus momentum: filled δx , open $-2T$, curves the parameter-free Fano prediction. (c) Measured reflection against $R = [1 + (4\omega v/J_1^2)^2]^{-1}$, $\omega = J(\Delta - \cos k)$, $v = J \sin k$: collapse over $\Delta \in [1.5, 12]$, $k_0 \in [0.4, 1.8]$ to 0.9–5.8%; as $k_0 \rightarrow 0$ transmission vanishes quadratically, Eq. (3): a soft magnon leaves no memory. Inset: the spin- s falsifier of Conjecture B—soft phase slope (filled, rings up to 480 sites) and memory quantum $-\delta x/\langle N_T \rangle$ (open: every physical wavepacket run with trapped weight $< 10^{-2}$, the record’s quality gate; controls excluded) against the line $1/s$, for $s = \frac{1}{2}, 1, \frac{3}{2}, 2$ (memory: $s \leq \frac{3}{2}$). Both are consistent with $1/s$, excluding the constant 2; that both measure one charge datum $|q_{\text{hard}}|/s$ is Conjecture B.

structural parallel to the celestial program’s symmetry towers at the boundary of spacetime [1]; its symplectic origin is deferred to forthcoming work. The wall is a broken asymptotic symmetry made visible, and—if Conjecture B holds—the magnon’s soft phase and the wall’s memory read out one conserved charge crossing an observer. The infrared structure of gauge theory and grav-

ity, long tied to null infinity, fits on a chain of spins.

We thank D. Giulini for discussions.

Corner A: the asymptotic charge algebra

Ground truth: repository shards `theory/corner-a.md`, `corner-a-kinks.md`, `corner-a-goldstone.md` (claim rows WI, A1, A2, G0, all PROVED through the adversarial verification loop recorded there); numerical certificates in Appendix E. Throughout, α labels vacua, $H_\alpha = \{g : g \cdot \alpha = \alpha\}$ the unbroken subgroup, and phases are normal-ordered so that the extensive phase of the truncated symmetry is removed.

Truncated symmetry (WI). For a finite region $R = [a, b]$ and any $g \in G$, exactly and with no limit, $U_R(g) |\psi(A_\alpha)\rangle = e^{i|R|\theta_\alpha(g)} |\psi(T_R^{(g)})\rangle$, where $T_R^{(g)}$ carries A_α outside R , $A_{g \cdot \alpha}$ inside, and two bond insertions: $V_\alpha(g)^{-1}$ on $\partial_- R$ and $V_\alpha(g)$ on $\partial_+ R$. The window must contain one site beyond each cut (or carry explicit edge-bond insertions); with interior bonds only, the identity is false, and the orientation of the two insertions matters—both failure modes are exhibited numerically in the certificate.

Unbroken case (A1). (a) Half-infinite strings act on states exactly as single bond insertions, and the convergence is eventually constant on every local algebra, not asymptotic. (b) They are implemented by a strongly convergent operator sequence if and only if $V_\alpha(g)$ is scalar. (c) Two insertions give the same state iff they differ by a scalar, so the endpoint states form a $PGL(\chi)$ -torsor. (d) On windows padded about the bond the map $M \mapsto$ insertion is injective, and the insertions realize a linear representation of the twisted group algebra $\mathbb{C}_{\omega_\alpha}[G]$ in which the multiplier acts nontrivially; padding is necessary—an explicit $\mathbb{Z}_2$ -symmetric counterexample shows the unpadded action is ill defined. On states the multiplier dies and what remains is a homomorphism $G \rightarrow PGL(\chi)$; what $[\omega_\alpha]$ obstructs is lifting it to an honest unitary representation. (e) *Open (H-split)*: that these endpoint states are normal states of the physical half-infinite algebra—equivalently that $\mathbb{C}_\omega[G]$ acts on a physical edge Hilbert space—is not proved; the expected route is the split property. Every physical-edge statement in this Letter carries this hypothesis explicitly.

Broken case (A2). For $g \notin H_\alpha$, every finite truncation stays in the vacuum sector (a kink-antikink pair at the two ends); the half-infinite limit exists in the weak-* sense only, with an exponential rate, and lands in the kink sector $\mathcal{K}_{\alpha, g \cdot \alpha}$, disjoint from the vacuum folium. The jump is invisible at every finite truncation and created entirely by the surviving end. Under the transitivity hypothesis (T), vacuum pairs form $(G/H_\alpha) \times (G/H_\alpha)$ and the complete diagonal invariant is the double coset in $H_\alpha \backslash G/H_\alpha$; for $G = SU(2)$ broken to $U(1)$ this is the relative angle of the two vacuum axes. Without (T) the

classification holds per orbit. Two load-bearing gaps are recorded in the shards and respected here: H-split above, and the absence of a g -uniform statement when the vacuum manifold is continuous.

Currents (G0). For any generator ξ and finite-range invariant H : $[H, q_x(\xi)] = j_{x-1|x}(\xi) - j_{x|x+1}(\xi)$ with the local cut current, and in wavepacket form $[H, Q_k] = (e^{ik} - 1)J_k$. Separately, for a normal-ordered *unbroken* generator $\xi \in \mathfrak{h}_\alpha$ only: on the vacuum, the charge density acts as the lattice divergence of a virtual bond quantity—the potential is the fundamental object, as a theorem. The finite-window Fourier identity carries two $\Theta(1)$ boundary terms; the clean $(1 - e^{ik})$ form holds only in the stated δ -normalized sense. The kinematic factor supplies no soft theorem (a withdrawn overclaim is recorded as a refuted row in the claims ledger).

Corner C: the two-body soft theorem and its limits

Ground truth: `theory/soft-current-recon.md` (row S2-2body, PROVED), `theory/oracle-bethe.md` (oracle facts O1–O10, verification role only), `theory/ml2-completeness.md` (row ML2, PROVED), `theory/spin-s-twomagnon.md` (row S2-2body-S, PROVED through its adversarial verification round), `theory/ml4-ward-reduction.md` and `theory/ml5-universality.md` (rows ML4-A, ML4-Ward, ML5-A/B PROVED as conditional implications; ML5 unrestricted REFUTED).

Derivation of Theorem 1. The exact current of the model is $j_{x,x+1}^- = \frac{J}{2}(S_{x+1}^- - S_x^-)P_{x,x+1}$. A two-magnon eigenwave is free away from contact; writing the incoming coefficient as 1 and the outgoing as $s(k_s, k_h)$, the single contact bond imposes

$$(2z_h - z_s z_h - 1)s + (2z_s - z_s z_h - 1) = 0, \quad z = e^{ik}, \quad (4)$$

whose unique analytic solution near $k_s = 0$ has k_s -coefficient $2i$, all hard dependence cancelling; the continuous logarithm gives Eq. (1), with remainder uniform on compact hard sets. The Ward reading: the state $Q_0 |k_h\rangle$ is the Bethe descendant with one rapidity at infinity, and $\langle k_h | Q_0^\dagger J_0^- | k_h \rangle = 2iv_h$; dividing a hard external-leg reduction by the energy shift $v_h k_s$ cancels the velocity and leaves the 2. Symmetry alone does not prove the theorem: the current has an orthogonal contact component, and what makes it subleading in the two-body sector is energy-shell matching plus C^1 regularity of the on-shell trace (an abstract cancellation lemma proved in the shard), not the Ward identity. Completeness of the two-magnon resolution—scattering states, one bound band, descendants, and one singular vector on even rings—is proved by direct Jacobi diagonalization; no Bethe completeness is assumed, and integrability appears nowhere in the hypotheses.

The universality boundary. With the four-site local operator $D = S_0^- S_1^- - S_1^- S_2^- + S_2^- S_3^- - S_3^- S_4^-$, the source $O_\eta = S_0^- + \eta D$ creates the same one-magnon amplitude as S_0^- but shifts the linear soft coefficient of the induced two-body amplitude by $2i\eta(1 - e^{-3ik_h})$. Unrestricted process independence is therefore refuted. The repaired statement (proved as a conditional implication) holds on the source class $\mathcal{S}_W$ defined by five conditions—an exhaustive normed LSZ decomposition, Ward covariance of the descendant current residue, a process-independent C^1 kinematic LSZ normalization, reduced-channel regularity with relative norm control, and no direct soft contact—and Conjecture S, in the form $M_{n+1} = S(k_s)M_n + o(k_s)$ with S a function of the asymptotic leg charges and velocities only, asserts universality on exactly this class. Open, and stated as such: infinite-volume wave operators (ML1), uniform form-factor regularity including the $k = \Theta(1/N)$ regime (ML3), the packet-smeared infinite-volume one-hard-leg estimate and the $n \geq 2$ -hard channel (ML4; the fixed-volume formulas are an off-shell interpolation whose naive volume-uniform reading is refuted), the exhaustive LSZ decomposition itself, nonemptiness of and microscopic membership in $\mathcal{S}_W$, and the limit-order control (ML6).

Spin- s slope (theorem). For the ferromagnet $H_S = -J \sum_x (\mathbf{S}_x \cdot \mathbf{S}_{x+1} - S^2)$ at every site spin S , the separated, adjacent, and (for $S \geq 1$) double-occupancy equations directly determine the closed-form regular two-magnon amplitude, with no integrability or completeness assumption; for each fixed $0 < |k_h| < \pi$ the physical phase has $\partial_{k_s} \delta_{\text{phys}}|_0 = \text{sgn}(v_h - v_s)/S$, locally uniformly on compact hard subsets with fixed channel (row S2-2body-S, PROVED). The scope is exact: this proves the unit-charge two-body slope only—not endpoint or equal-velocity limits, not spin- S Bethe completeness, not Conjecture S, not the memory half, not the $|q_{\text{hard}}| > 1$ factor, and not Conjecture B.

Corner B: flux identity, memory law, Fano reduction

Ground truth: `theory/memory-quantization.md` (rows M-flux PROVED; M-quant PROVED conditional on the hypothesis D18, the scattering content of H-AD-G; Mq-AD3 PROVED conditional on the sector-reduction hypothesis Mq-E), `theory/memory-quantization-general.md` (row M-quant-G, PROVED as a conditional implication), `theory/corner-b-draft.md` (rows K1–K3, B3 PROVED; K4 CONJECTURE; row M REFUTED).

Flux identity (exact). For any state, any window $W = [a, b]$, and any finite times, $\frac{d}{dt} \mathfrak{X}_W$ equals $\frac{1}{2s}$ times the difference of the two boundary-bond currents; integrating, $\delta x = \frac{1}{2s} [\tilde{j}_{a-1|a}(0) - \tilde{j}_{b|b+1}(0)]$, the zero-frequency weight of the physical boundary current. No kink, spectral, or asymptotic hypothesis enters. (An earlier reading

of this identity through virtual bond data was deleted during verification; the current here is physical.)

Hypothesis H-AD-G. For one fixed normalizable kink-magnon wavepacket, wave operators $W_\pm$ exist and are complete on its sector; the outgoing space splits into exactly one reflected and one transmitted channel with definite charges $q_{\text{in}} = q_L = -1$, $q_T = +1$, and no further propagating channel; the packet has no bound-state component; local observables decay on the moving legs; and limits are taken in the stated order (infinite volume, then time, then window). For Theorem 2 in the general register these charges, the density jump $2s$, and summable charge deviations from the two vacua (an ℓ^1 kink class) are the only inputs: conservation of the regularized charge $2s(\mathfrak{X}_W - c)$ plus leg bookkeeping gives $2s \delta x + (q_{\text{out}} - q_{\text{in}}) = 0$ per channel, hence Eq. (2); the quantum for a charge- ν transfer is $-\nu/2s$. The structural remark in the text is the shard’s: covariance forces $\omega_{g \cdot \alpha}(\xi) = \omega_\alpha(\text{Ad}_{g^{-1}} \xi)$, so if the broken element commutes with the unbroken direction ($\text{Ad}_{g^{-1}} \xi = \xi$, as for any abelian G) the two vacua carry *equal* densities and no such kink pair exists; the required inversion is supplied by a Weyl-type element, as the spin flip in XXZ. Nothing in compactness, injectivity, or finite range implies H-AD-G itself; it is proved here only where stated next.

Sector reduction and the Fano level. In the XXZ chain at $s = \frac{1}{2}$ the incoming component with at most three domain walls, under the enumeration hypothesis Mq-E (verified explicitly at $N = 14$; all-volume form open), is unitarily a two-sided tight-binding chain with one side-coupled level—the flat kink state, whose bondwise annihilation by the exact kink Hamiltonian is proved. Kato–Rosenblum and a Feshbach analysis then give H-AD-G’s scattering clauses for this reduced dynamics: trace-class perturbation, complete wave operators, no singular-continuous spectrum. Eliminating the side level yields $t(k) = [1 + iJ^2/4\omega v]^{-1}$ and Eq. (3). Leakage of the full chain out of the reduced sector is measured at $O(\Delta^{-2})$ (probabilities 0.10/0.03/0.008 at $\Delta = 2/4/8$); it affects $T(k)$, not the conservation laws. Full-chain H-AD-G and the universality of the soft-zero coefficient remain open and are tracked as such in the claims ledger.

Refutation of the naive dictionary. The brief’s Conjecture M (“ δx is the DC limit of the soft factor”) fails on two counts: the channel quantum is independent of the scattering phase once the channels exist, and a phaseless purely transmitting wall still displaces maximally. The refuted row is retained in the ledger with its counterexamples; the surviving pair is the flux identity plus Eq. (2).

Label rigidity. Under stationary vacua and finite-range (Lieb–Robinson) dynamics the factorized vacuum-pair boundary label of the kink sector is preserved at all finite times, so δx is motion within a fixed sector, not a change of sector. The thermodynamic uniqueness/flatness of the

kink family (a $\mathbb{Z}$ -torsor reading) remains a conjecture with finite-volume evidence.

The SPT rigidity dichotomy

Ground truth: `theory/spt-rebuild.md` (rows SPT-B-mult, SPT-B', SPT-E-AKLT, SPT-E', SPT-T', SPT-D', SPT-M' all PROVED with the hypotheses below explicit; SPT-M'-dyn CONJECTURE; the earlier point-wise bulk-blindness claim and the all-orders edge no-go both REFUTED and retained as negative rows).

Bulk (deformable). A closed on-site insertion leaves $V(g) \otimes \bar{V}(g)$: the multipliers cancel exactly, for every lift, so no closed-bulk amplitude carries a projective anomaly. This does not make bulk data blind to the phase—adjoint representation content distinguishes Pauli-projective from scalar conjugation—but it is deformable: along any common-gap symmetric path with continuous external data (tensors, channel embeddings, gauge fixes, normalizations), every normalized bulk coefficient is continuous; a coefficient reached through a soft limit, or through p derivatives, requires in addition C^p external data and the uniformity hypothesis H-soft- p of the shard. On the anisotropic AKLT family ($A_b^x \propto X$, $A_b^y \propto Y$, $A_b^z \propto Z$) the soft-charge coefficient is $C_{\text{bulk}}(b) = b^2/4(1 - b^2)$: $\frac{1}{8}$ at the AKLT point, 0.2402 at $b = 0.7$, same phase, same $[\omega]$.

Edge (rigid). On the fixed χ -dimensional Schmidt register of a half-chain, the compensated endpoint residue of a symmetry string is $V(g)$ itself, and the Hermitian charge residue is the centered generator $Q_{\text{edge}} = -iX^\circ$, lift-gauge invariant, with spectrum in the shifted lattice $q_\omega^\circ + \mathbb{Z}$ and every irreducible summand of dimension $\geq d_\omega$ (the minimal ω -projective dimension; > 1 whenever $[\omega] \neq 0$). For the AKLT family the finite-window residue is exactly $-\frac{1}{2}[1 - (2b^2 - 1)^L]Z$, with limit $-Z/2$ independent of b ; the trivial comparison tensor gives 0. The half/integer offset is pinned by the global group ($O(2)$ here, or the $SU(2)$ cover); a bare $U(1)$ does not protect it.

Edge memory. Given H-split, the edge-resolved scattering hypothesis H-AD-edge, charge conservation, and definite channel charges, $\Delta Q_{\text{edge}} = -(Q_{\text{bulk,out}} - Q_{\text{bulk,in}})$, with channel outcomes in $\mathbb{Z}$ (for the AKLT doublet, in $\{-1, 0, 1\}$)—the same bookkeeping as Theorem 2 with edge charge in place of wall position. Protection is capacity protection: the projective multiplet cannot be split symmetrically. That the specific open AKLT parent Hamiltonian with boundary coupling $P_{0,1}^{(S=2)}$ has a nonzero edge-changing reflection amplitude on an open momentum interval is conjecture SPT-M'-dyn; the deciding half-chain scattering computation, with pre-registered acceptance gates, is specified in the shard. A null result there refutes that conjecture only, not the registered endpoint theorems.

Numerical certificates

The theorems above carry independent machine-checked certificates under `theory/checks/` (ten checkers, all passing; five of them—the Ward-reduction/universality, flux/memory-scan, general-memory, spin- S slope, and SPT checkers—carry documented failing “red” mutation modes), and the quantitative claims of the text trace to committed data files: exact-diagonalization form-factor residuals $\leq 1.6 \times 10^{-14}$ and coefficient fits to 2.2×10^{-10} (`soft_current_recon_check.py`); two-magnon completeness on rings up to 26 sites (`m12_completeness_check.py`); the flux identity to 3.4×10^{-16} (`mquant_check.py`); the wavepacket memory scan (`memory-scan-1.json`: $N = 160$, sparse kink-sector propagation; $|\delta x + 2\langle N_T \rangle| \leq 0.064$ sites across the full momentum scan, dominated by the slowest packets with trapped weight $\approx 4 \times 10^{-3}$, and ≤ 0.005 sites for both estimators on the nine rows with trapped weight $< 10^{-6}$); the Fano cross-check (0.9–5.8% over $\Delta \in \{1.5, 2, 3, 6, 12\}$, pre-registered 8% criterion); the spin- s falsifier and its independent cross-check (`spin1-bc-falsifier.json`, `spin1-bc-crosscheck.json`: ansatz-free ring spectra to $N = 480$ plus real-time collisions, slope deviations from $1/s$ below 0.5% extrapolated; wavepacket memory ratios within 2.7% under the trapped-weight gate $< 10^{-2}$, within 1% at the cleanest kinematics); and the SPT deciding computation (`spt_rebuild_check.py`: bulk coefficient and edge residue to stated tolerances, sign and phase-gauge mutants failing). Figures 1–3 are generated from these committed files by `paper/figures/make_figures.py`; no number is hand-entered.

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